

Renewable Energy Study Notes

Overview

Renewable energy comes from sources that are naturally replenished, such as sunlight, wind, water, biomass, and geothermal heat. These sources are considered more sustainable than fossil fuels because they can be used repeatedly without being depleted on a human timescale.

Main Types of Renewable Energy

Solar Energy

Solar energy is generated by capturing sunlight using photovoltaic panels or solar thermal systems. Photovoltaic cells convert sunlight directly into electricity.

Wind Energy

Wind turbines convert the kinetic energy of moving air into electrical power. Wind farms may be built on land or offshore.

Hydropower

Hydropower uses flowing or falling water to spin turbines and generate electricity. Large dams and small run-of-river systems are common examples.

Biomass Energy

Biomass energy comes from organic material such as crop waste, wood, or animal matter. It can be burned directly or converted into biofuels.

Geothermal Energy

Geothermal systems use heat from inside the Earth to generate electricity or provide direct heating.

Advantages of Renewable Energy

- Lower greenhouse gas emissions compared with coal, oil, and natural gas.
- Reduced dependence on imported fossil fuels.
- Long-term sustainability.
- Potential for decentralized energy generation in rural and remote areas.

Challenges

- Solar and wind are intermittent because they depend on weather conditions.
- Energy storage systems such as batteries are often needed.
- Initial infrastructure costs can be high.
- Some hydropower projects can affect ecosystems and local communities.

Key Terms

- **Photovoltaic cell:** A device that converts sunlight directly into electricity.
- **Intermittency:** Variability in energy output due to changing natural conditions.
- **Grid integration:** The process of connecting power generation sources to the electricity grid.
- **Energy storage:** Methods used to store electricity for later use.

Short Questions

1. What is renewable energy?
2. How do photovoltaic panels generate electricity?
3. Why is wind energy considered intermittent?
4. Name two advantages of hydropower.
5. What are the environmental concerns linked to some hydropower projects?

Practice Notes

Countries are investing in renewable energy to reduce emissions and improve energy security. A balanced energy system often combines multiple renewable sources along with storage and transmission upgrades.